UNIVERSITY SCHOLARSHIP ALLOCATION

Python Program – One Student

```
# University Scholarship Allocation (One Student)

name = input("Enter student name: ")

aggregate = float(input("Enter aggregate marks: "))
sub1 = float(input("Enter Subject 1 marks: "))
sub2 = float(input("Enter Subject 2 marks: "))
sub3 = float(input("Enter Subject 3 marks: "))

attendance = float(input("Enter attendance percentage: "))
income = float(input("Enter family income: "))

if (aggregate >= 75 and
    sub1 >= 50 and
    sub2 >= 50 and
    sub3 >= 50 and
    attendance >= 80 and
    income <= 300000):

    print("\\nScholarship Approved")
    print("Name:", name)

else:
    print("\\nScholarship Not Approved")
```

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Assignment Matter – One Student

1. Problem Statement

Develop a Python program to check whether a student is eligible for a university scholarship based on aggregate marks, marks in three subjects, attendance percentage, and family income.

2. Eligibility Conditions

• Aggregate marks must be 75 or above.
• Marks in each of the three subjects must be 50 or above.
• Attendance must be 80% or above.
• Family income must be Rs. 3,00,000 or below.

3. Algorithm

1. Start the program.
2. Read the student's name.
3. Read aggregate marks and marks in three subjects.
4. Read attendance percentage and family income.
5. Check whether all scholarship eligibility conditions are satisfied.
6. If all conditions are true, display “Scholarship Approved” and the student's details.
7. Otherwise, display “Scholarship Not Approved”.
8. Stop the program.

4. Pseudocode

```
START
INPUT name
INPUT aggregate
INPUT sub1, sub2, sub3
INPUT attendance
INPUT income

IF aggregate >= 75 AND
   sub1 >= 50 AND
   sub2 >= 50 AND
   sub3 >= 50 AND
   attendance >= 80 AND
   income <= 300000 THEN
       PRINT "Scholarship Approved"
       PRINT name
ELSE
       PRINT "Scholarship Not Approved"
       PRINT name
END IF

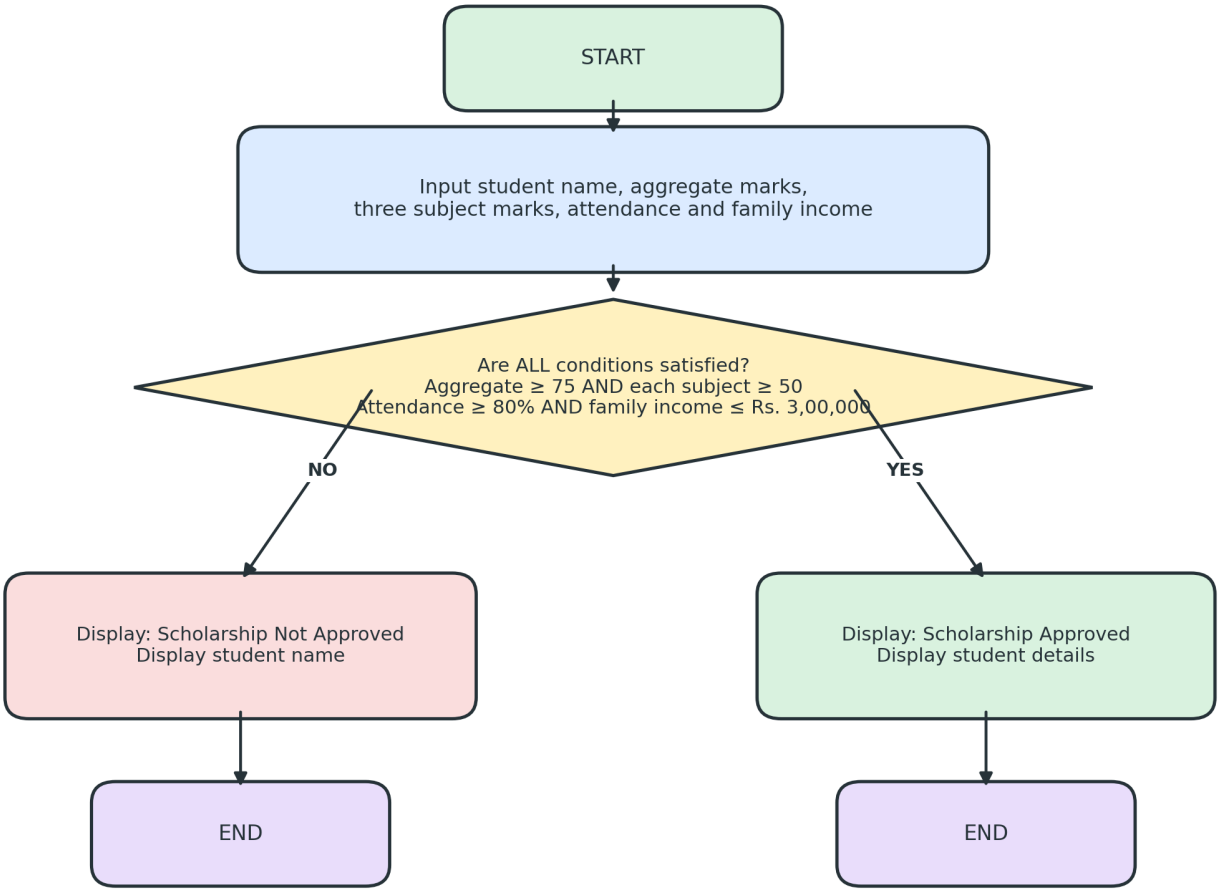
STOP
```

5. Flowchart

The flowchart below shows the input, eligibility decision, and both possible results.

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Flowchart – One Student



FLOWCHART SYMBOLS



Note: The student is approved only when every eligibility condition is true.

6. Sample Output – Scholarship Approved

```
Enter student name: Bharath
Enter aggregate marks: 85
Enter Subject 1 marks: 75
Enter Subject 2 marks: 80
Enter Subject 3 marks: 90
Enter attendance percentage: 90
Enter family income: 200000

----- Scholarship Result -----
Scholarship Approved
Student Name: Bharath
Aggregate Marks: 85.0
Attendance: 90.0 %
Family Income: Rs. 200000.0
```

7. Python Concepts Used

• Variables: name, aggregate, sub1, sub2, sub3, attendance, income.
• Data types: string and float.
• Input and output: input() and print().
• Operators: comparison operators (>=, <=) and logical operator (and).
• Conditional statements: if and else.

Conclusion: The program approves the scholarship only when every eligibility condition is satisfied. If even one condition is not satisfied, the scholarship is not approved.